

Question number	Answer	Additional guidance	Mark
3(a)	A description that includes four of the following points: - vaccine contains an {inactive / attenuated} form of the {pathogen / virus / bacteria / microorganism} (1) - macrophages engulf and display antigen on cell surface (1) - macrophages will present the antigen to T helper cells (1) - T helper cells will activate {B / T killer} cells (1) (T / B) memory cells produced (1)	ACCEPT antigen (found on the pathogen) / weakened pathogen DO NOT ACCEPT dead {virus / antigen} ACCEPT macrophages become antigen-presenting cells + description to T helper cells ACCEPT description e.g. T helper cells release cytokines that cause B cell proliferation	(4)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A description that includes the following points: • conclusion relating trust in scientists to income (1) • conclusion relating trust in scientists to vaccine safety (1) • conclusion relating income to vaccine safety (1)	e.g. trust goes down (slightly) with increase in income e.g. people who agree that vaccines are safe trust the scientists the most e.g. at all incomes, there are people who agree, neither agree nor disagree and disagree that vaccines are safe	(3)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	education / (influence of) {media / family / friends / internet} / religious beliefs	previous experiences with {vaccines / medications} ethical beliefs fear of needles lack of trust in testing process allergies side effects more money for {treatment / healthcare} medical insurance	(1)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes two of the following points: - vaccinated people will {not / be less likely to} develop the disease / vaccinated people will be immune (1) - fewer infected people to infect people who are not immune (1) - protect people who cannot become immune (1)	IGNORE less likely to be infected ACCEPT herd immunity e.g. allergic to vaccines, immunodeficient, immunosuppressed	(2)

